

Ujjwal Pawar

PhD, University of Edinburgh

*Room 1.17, Informatics Forum, 10 Crichton Street
Edinburgh EH8 9AB, UK
H +44(0)7404958297
B u.pawar@sms.ed.ac.uk
LinkedIn ■ Website*

Professional Summary

Ph.D. researcher in 5G/6G systems with hands-on experience building and operating large-scale multi-vendor Open RAN testbeds, specializing in RAN optimization, telemetry, emulation, and AI-driven network automation.

Nationality

India

Education

- Oct 2022–Present **Ph.D. in Computer Science and Engineering**, *University of Edinburgh, School of Informatics*, Edinburgh, UK.
- Jan 2019–Dec 2021 **M.Tech in Computer Science and Engineering**, *Indian Institute of Technology, Hyderabad*, Hyderabad, India, CGPA 9.19/10.
Thesis: Traffic-Aware Compute Resource Tuning for Energy Efficient Cloud RANs.
- Aug 2014–May 2018 **Bachelor of Engineering in Computer Science and Engineering**, *University Institute of Technology, RGPV*, Bhopal, India, CGPA 7.3/10.

Experience

- Sep 2026–Present **AI/ML Research Intern – Open RAN & 6G Communications**, *Keysight Technologies*, Edinburgh, Scotland, UK.
- Jan 2025–Jun 2025 **Research Associate**, *Netsys Lab, School of Informatics, University of Edinburgh*, Edinburgh, UK.
Built a 20-cell private 5G network and currently manage its operation.
- Jan 2022–Sept 2022 **System Specification Engineer**, *Nokia R&D*, Bangalore, India.
Worked on Link Adaptation algorithms to improve VoNR performance.
- Jan 2019–Dec 2021 **Research Assistant**, *Indian Institute of Technology, Hyderabad*, Hyderabad, India.
Developed solutions for building energy-efficient data centers for 5G RAN.

Conference / Journal Papers

- **U. Pawar**, A. Ferguson, Y. Xia, X. Foukas, M. Marina, and B. Radunovic, “Chronos: Scalable and Cost-Effective Cloud-based RAN Emulation System,” ACM MobiCom, 2026.

- A. E. Ferguson, **U. Pawar**, T. Wang, and M. K. Marina, “Campus5G: A Campus Scale Private 5G Open RAN Testbed,” *ACM SIGCOMM Computer Communication Review*, Volume 55, Issue 3, pp. 19–28, 2025.
- C. Sun, **U. Pawar**, M. Khoja, X. Foukas, M. K. Marina, and B. Radunovic, “SpotLight: Accurate, Explainable and Efficient Anomaly Detection for Open RAN,” *ACM MobiCom*, 2024.
- M. Sharma, **U. Pawar**, A. Antony Franklin, and B. R. Tamma, “Proactive Clustering of Base Stations in 5G C-RAN using Cellular Traffic Prediction,” *IEEE 8th International Conference on Network Softwarization (NetSoft)*, pp. 339–347, 2022.
- **U. Pawar**, B. R. Tamma, and F. A. Antony, “Traffic-Aware Compute Resource Tuning for Energy Efficient Cloud RANs,” *IEEE Global Communications Conference (GLOBECOM)*, pp. 01–06, 2021.

Demo Papers

- **U. Pawar**, A. Ferguson, Y. Xia, X. Foukas, M. Marina, and B. Radunovic, “Chronos: A Cloud-Based RAN Emulation System,” *32nd Annual International Conference on Mobile Computing and Networking Demos (MobiCom '26 Demos)*, 2026.
- **U. Pawar**, A. Ferguson, T. Wang, and M. Marina, “A Campus Scale Private 5G Open RAN Testbed,” *ACM MobiCom Demo*, pp. 1248–1250, 2025.
- C. Sun, **U. Pawar**, M. Khoja, X. Foukas, M. K. Marina, and B. Radunovic, “SpotLight–An Open RAN Anomaly Detection and Identification System,” *ACM MobiCom Demo*, pp. 1769–1771, 2024.

Workshop Papers

- **U. Pawar**, J. Yang, A. Ferguson, D. Mahendiran, and M. Marina, “Panoramic: A Cross-Layer Measurement Platform for Open RAN Deployments,” *Proceedings of the 3rd ACM Workshop on Open and AI RAN (OpenRan '26)*, 2026.
- **U. Pawar**, A. E. Ferguson, Y. Takano, J. Larrea, X. Foukas, M. K. Marina, and B. Radunovic, “Towards Scalable and Cost-Effective RAN Emulation Leveraging the Public Cloud,” *ACM HotMobile*, pp. 43–48, 2025.

Thesis

- **U. Pawar** and B. R. Tamma, “Traffic-Aware Compute Resource Tuning of vRANs for Energy Efficient 5G Networks,” *Master’s Thesis*, 2021.

Awards and Scholarship

- 2026 **Travel Grant**, *ACM MobiCom '26*.
- Nov 2025 **Best Demo Award**, *ACM MobiCom '25*.
- Oct **PhD Studentship**, *School of Informatics, University of Edinburgh*.
- 2022–Aug
- 2026

Jan 2019–Dec 2021 **Research Assistant Studentship**, *Department of Computer Science, Indian Institute of Technology, Hyderabad.*

Leadership and Teamwork

2022, 2024, **Marker**, *School of Informatics, University of Edinburgh.*

2026 Marker for Computer Networks.

2020–2021 **Teaching Assistant**, *Department of Computer Science and Engineering, Indian Institute of Technology, Hyderabad.*

TA for Networked Wireless Systems (CS5070), Internet of Things (CS5453), Network Security (CS6903), and SDN (CS5643).

References

Name Mahesh K. Marina

Designation Professor, University of Edinburgh